

Auxiliary Material for “Probabilistic-Aware Actor-Critic Eco-ACC for Hybrid Electric Vehicles Utilizing Traffic Information”

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This auxiliary material is openly available, together with the source code, the processed car-following scenario files, the per-seed evaluation logs, and the evaluation pipeline, at <https://github.com/Dexter753/PA-CSAC-OpenSource>.

Abstract

This auxiliary material provides the complete parameter listings and full-length result tables underlying the results reported in the paper: per-step multi-horizon prediction performance (Table S1), the classification and values of all reward- and cost-related coefficients (Table S2), the complete parameter sets of the kinematics-based dynamic safety shield (Tables S3 and S4), the full prediction-error injection grid of the robustness analysis (Table S5), the per-seed raw values underlying every mean $\pm$ std entry of the main comparison table (Table S6), the per-seed raw values of the five-seed ablation configurations (Table S7), the per-seed raw values of the five-seed reward-coefficient sensitivity configurations (Table S8), the all-scenario unconditional statistics over the complete test set (Table S9), the split-conformal recalibration analysis of the deployed dispersion channel (Table S10), the per-seed numeric traces of the PPO-Lagrangian dual variable underlying the convergence analysis (Table S11), the per-seed raw values of the shield-off replay (Table S12), the per-scenario detail of the seed-42 validity-gate exclusions (Table S13), the per-seed raw values of the five-seed constraint-handling cross re-evaluated under the unified protocol (Table S14), the efficiency-robustness replay of the fuel accounting over all six perturbed efficiency settings (Table S15), the per-seed raw values of the dispersion-source substitution replay (Table S16), the shield-off replay of the five learning baselines (Table S17), the closed-form expressions of the auxiliary reward terms and the near-boundary safety correction (Table S18), the complete per-method computational-cost listing (Table S19), the complete baseline configurations and shared training hyperparameters (Section S19), the vehicle and energy-model parameter configuration (Section S20 and Table S20), the complete reward-coefficient sensitivity screening grid (Section S21 and Table S21), the probabilistic calibration quality analysis (Section S22 and Table S22), the checkpoint-independence audit of the PPO-Lagrangian seeds 52 and 62 (Section S23 and Tables S23 and S24), and the conditional basis of the speed- and time-normalised comparison (Section S24 and Table S25); the reliability diagram of the calibrated predictive distribution is reported in the main text.

S1. MULTI-STEP PREDICTION PERFORMANCE

Table S1 reports the per-step prediction performance of the adopted Transformer + TFG + GMH model over the 10-step (10 s) prediction horizon, evaluated on the held-out test split of the NGSIM US101 dataset. The leading-vehicle speed channel (Velocity) and the macroscopic flow speed channel (Flow) are listed separately. The R^2 of the velocity channel decays gradually from 0.941 at step $t+1$ to 0.914 at step $t+10$, while the RMSE increases from 0.770 m/s to 0.916 m/s; the degradation of the flow speed channel over the same horizon is milder. The final row reports the horizon-aggregated metrics, computed as the arithmetic mean of the ten per-step entries; the deployment-level figures quoted in the main text (velocity-channel RMSE of 0.857 m/s and R^2 of 0.926) are these aggregated values.

S1.1 Architecture and Training Configuration of the Deployed Predictor

The deployed prediction model is the Transformer + TFG + GMH configuration selected by the component-screening search reported in the main text. Its concrete instance uses a model dimension of $d_{\text{model}} = 128$, one Transformer encoder layer with 2 attention heads, and a dropout rate of 0.195; the temporal feature gating (TFG) block and the Gaussian-mixture head (GMH) are both active, with the GMH decoding a three-component Gaussian mixture ($M = 3$) per predicted step. Training uses the Adam optimizer with a learning rate of 1.08×10^{-4} and a weight decay of 1.48×10^{-4} . The input window, causal separation interval, and output horizon follow the values declared in the main text ($N_h = 10$, $N_g = 5$, $N_p = 10$). The per-epoch training loss of the selected configuration, together with the loss traces of the component-screening seeds, is recorded in the released training monitors

TABLE S1
PER-STEP PREDICTION PERFORMANCE OVER THE 10-STEP HORIZON (TEST SPLIT)

Step	Leading vehicle speed			Macroscopic flow speed		
	RMSE (m/s)	MAE (m/s)	R^2	RMSE (m/s)	MAE (m/s)	R^2
$t + 1$	0.770	0.573	0.941	0.610	0.428	0.907
$t + 2$	0.788	0.620	0.938	0.634	0.456	0.900
$t + 3$	0.776	0.618	0.940	0.650	0.467	0.895
$t + 4$	0.880	0.693	0.923	0.700	0.508	0.878
$t + 5$	0.837	0.667	0.930	0.633	0.463	0.901
$t + 6$	0.851	0.679	0.927	0.633	0.459	0.901
$t + 7$	0.881	0.698	0.922	0.649	0.471	0.896
$t + 8$	0.930	0.730	0.912	0.632	0.460	0.901
$t + 9$	0.945	0.740	0.909	0.597	0.438	0.912
$t + 10$	0.916	0.719	0.914	0.609	0.448	0.908
Horizon-aggregated	0.857	0.674	0.926	0.635	0.460	0.900

(training_performance_monitor_*.csv of the accompanying repository), so the convergence behavior of the predictor is auditable without an additional figure.

S1.2 Precise Definitions of the 8-Dimensional Derived Probabilistic Features

The 8-dimensional derived probabilistic feature group of the state space is produced by a fixed deterministic mapping of a 9-dimensional raw predictive vector $\mathbf{p} = [\mu_1, \mu_2, \mu_3, \sigma_1, \sigma_2, \sigma_3, v_{lb}, v_{ub}, \rho_p]$, where μ_i and σ_i ($i = 1, 2, 3$) denote the predicted mean and standard deviation of the leading-vehicle speed at three selected future steps, $[v_{lb}, v_{ub}]$ is the 95% prediction interval, and ρ_p is the traffic flow density attached to the prediction. With $\bar{\mu} = \frac{1}{3} \sum_i \mu_i$, $\bar{\sigma} = \frac{1}{3} \sum_i \sigma_i$, $w_{ci} = \max(v_{ub} - v_{lb}, 0)$, and clip denoting truncation to the stated bounds, the eight features are

$$\begin{aligned}
d_i &= \text{clip}\left(\frac{\mu_i - \bar{\mu}}{10}, -3, 3\right), \quad i = 1, 2, 3 && \text{(cross-step mean deviation, m/s scale)} \\
\sigma_n &= \text{clip}\left(\frac{\bar{\sigma}}{5}, 0, 3\right) && \text{(normalized dispersion)} \\
w_n &= \text{clip}\left(\frac{w_{ci}}{10}, 0, 6\right) && \text{(normalized interval width)} \\
\rho_n &= \text{clip}\left(\frac{\rho_p}{300}, 0, 3\right) && \text{(normalized density)} \\
r_{\text{pred}} &= \text{clip}\left(1 - \frac{\sigma_n}{3}, 0, 1\right) && \text{(prediction reliability)} \\
c_{\text{pred}} &= \text{clip}\left(1 - \frac{\text{std}(d_1, d_2, d_3)}{2}, 0, 1\right) && \text{(cross-step consistency),}
\end{aligned}$$

so that $\mathbf{x}_{\text{prob}} = [d_1, d_2, d_3, \sigma_n, w_n, \rho_n, r_{\text{pred}}, c_{\text{pred}}]^\top$ forms the input of the probabilistic feature embedding network. Reliability decreases monotonically with the normalized dispersion, and consistency measures the spread of the three per-step deviations from their common mean, i.e., whether the predicted trajectory drifts unevenly across its future steps.

S2. CLASSIFICATION OF REWARD- AND COST-RELATED COEFFICIENTS

Table S2 classifies every coefficient involved in the reward function, the safety cost function, and the adaptive weight mechanism of the methodology section in the main text into three categories: physically derived constants, scale and normalization constants, and behavioral weights. Only the third category involves manual calibration; all values remain fixed across the 17 test scenarios and all random seeds.

The four risk sub-terms entering the comprehensive risk coefficient ξ of Eq. (16) are normalized mappings from

TABLE S2
COMPLETE CLASSIFICATION AND VALUES OF THE REWARD- AND COST-RELATED COEFFICIENTS

Coefficient (symbol)	Value / derivation
<i>(a) Physically derived constants</i>	
Standstill gap d_0	5.0 m, minimum margin at standstill
Reaction headway τ_h	1.2 s
Comfortable deceleration $b_{\max}$	4.0 m/s ²
Uncertainty sensitivity k_{uncert}	0.9, matched to the predictor output scale
Ideal-zone ratio γ_s	1.2
Jerk limit	3.0 m/s ³
<i>(b) Scale and normalization constants</i>	
Energy normalization E_{norm}	8000 J
Reward bias b_r	0.15
Reward clipping range	$[-10, b_r + 0.05]$
Uncertainty reference σ_{ref}	1.8 m/s
Safety-zone widths $\beta_{s1}, \beta_{s2}, \beta_{s3}$	2.0, 3.0, 10.0 m
Following widths $\beta_{\text{far}}, \beta_{\text{close}}$	7.6, 3.6 m
Target band half-width	3.2 m
Near-boundary margin	2.0 m
Dangerous-zone deficit cap	10 m
Soft upper bound d_{up}	$\max(1.45 d^* + 3, 0.45 d_{\max}), d_{\max} = 120$ m
tanh inner scales (auxiliary terms)	4.5, 12.0, 9.5, 1.5, 3.2, 6.0, 2.5, 2.0, 0.8, 1.4, 2.2, 0.018
Cost saturation	$\min\{1, \cdot\}$ per step
Target-gap uncertainty scale κ_1	0.55 (converts predicted velocity std to distance increment at $\Delta t = 1$ s)
Target-gap constant offset κ_2	0.3 m
Risk normalization: gap-risk cap, TTC threshold, σ reference, SOC tolerance	1.5 d_{safe} , 5.0 s, 1.8 m/s, 0.30
TTC numerical guard ϵ	10^{-6} m/s (prevents division by zero)
<i>(c) Behavioral weights and calibrated trigger ratios (manually calibrated)</i>	
Lower-bound tolerance ratio	0.94 of d_{safe}
Hard upper bound	$\max(1.40 d_{\text{safe}}, 0.60 d_{\max})$
Primary coupling factor	0.95 in $r = 0.95(\omega_e r_e + \omega_s r_s) + \dots$
Energy reward α_e	0.8
Safety segments $\alpha_{s1}, \alpha_{s2}, \alpha_{s3}$	0.8, 0.1, 0.08
Near-boundary closing penalty	0.35, scales 0.8 and 0.6
Following asymmetry $\alpha_{\text{far}}, \alpha_{\text{close}}$	0.15, 0.42
In-band bonus amplitude	0.18
Gap tail penalty (following reward)	0.006 per m beyond $d^* + 10$ m; a weak anti-drift tail inside the following reward, distinct from the soft-upper-bound term below, which activates only beyond d_{up}
r_{gap} coefficients (beyond d_{up})	0.27 (tanh), 0.006 (linear), 2×10^{-4} (quadratic)
r_{vm}	0.18 with gate $0.35 + 0.65 \kappa_{\text{catch}}$
r_{stop}	0.40
r_{catch}	0.35
r_{bb}	0.12
r_{app}	0.12 and 0.08
r_{comf}	0.20 (jerk), 0.12 (acceleration)
r_{brake}	0.06
r_{soc}	0.14
r_{shield}	0.12 (push), 0.07 (pull), scale 0.8
Cost weights $c(s, a)$	upper 0.40, lower 0.35, jerk 0.12, shield c_{sh} 0.18/0.08 (scale 0.9)
Actor-side shield regularizer k_{sm}	0.18, pull component weighted 0.6 (actor objective of the main text)
Fixed penalty multiplier λ_{pen}	1.0
Cost threshold C_{lim}	0.30
Dynamic weight aggregation	$0.5 (0.7 r_{\text{gap}}^{\text{risk}} + 0.3 r_{\text{ttc}}^{\text{risk}}) + 0.3 r_{\sigma} + 0.2 r_{\text{soc}}$
Dynamic weight range	$\omega_s \in [0.40, 0.70]$, outer clip $[0.25, 0.75]$

Note: two distinct spacing thresholds appear in the reward. The gap tail penalty acts inside the following reward as a mild linear charge beyond $d^* + 10$ m, whereas the soft upper bound $d_{\text{up}} = \max(1.45 d^* + 3, 0.45 d_{\max})$ of panel (b) activates the separate r_{gap} term (tanh + linear + quadratic) only for persistent drop-out following; the two terms are cumulative by design.

physical observables to the unit interval:

$$\begin{aligned}
r_{\text{gap}}^{\text{risk}} &= 1 - \frac{\min(\Delta s / d_{\text{safe}}, 1.5)}{1.5}, \\
r_{\text{ttc}}^{\text{risk}} &= \max\left(0, 1 - \frac{\tau_{\text{ttc}}}{5.0 \text{ s}}\right), \\
r_{\sigma} &= \min\left(\frac{\hat{\sigma}}{\sigma_{\text{ref}}}, 1\right), \quad \sigma_{\text{ref}} = 1.8 \text{ m/s}, \\
r_{\text{soc}} &= \min\left(\frac{|\text{SOC} - \text{SOC}^*|}{0.30}, 1\right), \quad \text{SOC}^* = 0.60,
\end{aligned}$$

where Δs is the inter-vehicle gap, $\tau_{\text{ttc}} = \Delta s / ([v_e - v_l]_+ + \epsilon)$ is the time-to-collision with numerical guard $\epsilon = 10^{-6}$ m/s, and $\hat{\sigma}$ is the horizon-averaged predicted standard deviation. Each sub-term maps its argument monotonically into $[0, 1]$ before the weighted aggregation of Eq. (16).

S2.1 Complete Definitions of the Per-Step Safety Cost

The per-step safety cost of the main text aggregates discrete hard-violation events and continuous soft costs. Its indicators and soft costs are defined in full as follows:

$$\begin{aligned}
\mathbb{I}_{\text{hv}} &= \mathbb{I}[\Delta s < 0.94 d_{\text{safe}} \vee |j| > 3.0 \text{ m/s}^3 \vee \tau_{\text{ttc}} < 1.0 \text{ s}], \\
\mathbb{I}_{\text{up}} &= \mathbb{I}[\Delta s > \max(1.40 d_{\text{safe}}, 0.60 d_{\text{max}})], \\
c_{\text{lo}} &= \min\left\{1, \frac{(0.94 d_{\text{safe}} - \Delta s)_+}{\max(0.20 d_{\text{safe}}, 1.0)}\right\}, \\
c_{\text{jk}} &= \min\left\{1, \frac{(|j| - 2.55 \text{ m/s}^3)_+}{1.05 \text{ m/s}^3}\right\}, \\
c_{\text{sh}} &= \min\left\{1, 0.18 \tanh\left(\frac{\Delta a_{\text{push}}}{0.9}\right) + 0.08 \tanh\left(\frac{\Delta a_{\text{pull}}}{0.9}\right)\right\},
\end{aligned}$$

where $\mathbb{I}_{\text{hv}}$ is the hard-violation indicator covering lower-bound gap intrusion, excessive jerk, and TTC emergency; $\mathbb{I}_{\text{up}}$ is the soft upper-bound indicator; c_{lo} and c_{jk} are normalized soft costs quantifying the severity of the lower-bound shortfall and the jerk excess; and c_{sh} is the shield mismatch cost built from the downward and upward action modifications Δa_{push} and Δa_{pull} imposed by the shield, with $d_{\text{max}} = 120$ m the drop-out gap bound.

S2.2 Intermediate Quantities of the Auxiliary Reward Terms

The closed-form expressions of the auxiliary reward terms in the main text use the following intermediate quantities: $\Delta = \Delta s - d^*$ (gap deviation from the target), $e_{\Delta} = (\Delta s - d_{\text{up}})_+$ (excess beyond the soft upper bound), $\kappa_{\text{catch}} = \tanh((\Delta - 3.2)_+ / 6.0)$, $f_{\text{gap}} = \tanh(e_{\Delta} / 12.0)$, $f_{\text{rec}} = \tanh((\Delta - 3.2)_+ / 9.5)$, $f_v = \tanh((v_l - v_e) / 1.5)$, $d_{\text{close}} = (d^* - \Delta s)_+$, and $v_{\text{close}} = (v_e - v_l)_+$. Units: all gap-type quantities (Δ , Δs , d^* , d_{up} , e_{Δ} , d_{close}) are in meters; all velocities (v_e , v_l , v_{close}) in m/s; the acceleration a in m/s²; the jerk j in m/s³; Δa_{push} and Δa_{pull} in m/s²; and SOC is dimensionless. Every reward term is dimensionless (per-step reward units); the numerical coefficients inside tanh arguments carry the corresponding physical units (e.g., 4.5 in $e_{\Delta} / 4.5$ is in meters).

S3. COMPLETE PARAMETER LISTING OF THE DYNAMIC SAFETY SHIELD

Tables S3 and S4 list the complete parameter sets of the braking-side intervention rules and the active clearance control rules of the kinematics-based dynamic safety shield. The rule tables in the main text inline the threshold values; the listings here also document the action bounds and the absolute-gap floors that activate together with the ratio-based conditions.

S4. FULL RESULTS OF THE PREDICTION-ERROR INJECTION EXPERIMENT

Table S5 documents the complete 17-configuration prediction-error injection grid replayed on the converged checkpoints of all five random seeds (22, 32, 42, 52, 62) under the unified closed-loop protocol of the main comparison;

TABLE S3
COMPLETE PARAMETER SET OF THE BRAKING-SIDE INTERVENTION RULES (GAP-RATIO TRIGGER VALUES ARE CALIBRATED RATHER THAN DERIVED; THE 0.98 PREDICTED-GAP TRIGGER CORRESPONDS TO ONE STEP OF LOOK-AHEAD PROJECTION)

Parameter	Value
Action lower bound a_{low}	-3.0 m/s^2
Action upper bound a_{high}	2.0 m/s^2
Hard braking command	-2.2 m/s^2
Moderate braking command	-1.2 m/s^2
Light braking command	-0.8 m/s^2
Predicted-gap trigger ratio	$0.98 d_{\text{safe}}$ (one-step look-ahead)
Fast-closing relative speed trigger	$\Delta v < -1.2 \text{ m/s}$
Fast-closing gap ratio	$1.12 d_{\text{safe}}$
Moderate-closing relative speed trigger	$\Delta v < -0.3 \text{ m/s}$
Moderate-closing gap ratio	$1.05 d_{\text{safe}}$
TTC critical threshold	$1.0 \text{ s} \rightarrow \text{command } -3.0 \text{ m/s}^2$
TTC warning threshold	$1.8 \text{ s} \rightarrow \text{command } -2.4 \text{ m/s}^2$

TABLE S4
COMPLETE PARAMETER SET OF THE ACTIVE CLEARANCE CONTROL RULES

Parameter	Value
Rapid push: gap condition	$\Delta s > \max(1.40 d_{\text{safe}}, 35.0 \text{ m})$
Rapid push: relative speed	$\Delta v > 0.3 \text{ m/s}$
Rapid push action floor	$a \geq 0.0 \text{ m/s}^2$
Light forward push: gap condition	$\Delta s > \max(1.35 d_{\text{safe}}, 30.0 \text{ m})$
Light forward push: relative speed	$\Delta v > 0.5 \text{ m/s}$
Light forward push action floor	$a \geq -0.1 \text{ m/s}^2$
Moderate acceleration: gap condition	$\Delta s > \max(1.60 d_{\text{safe}}, 45.0 \text{ m})$
Moderate acceleration: relative speed	$\Delta v > 1.0 \text{ m/s}$
Moderate acceleration action floor	$a \geq 0.2 \text{ m/s}^2$

because the perturbations act on the prediction input at decision time, no retraining is involved. The robustness section of the main text reproduces the baseline and the representative gradient of each perturbation family; the full grid is tabulated here.

S5. PER-SEED RAW VALUES OF THE MAIN COMPARISON

Table S6 lists the per-seed raw values underlying every mean $\pm$ std entry of the quantitative comparison table in the main text for all learning-based methods (including the SMORL and HRL extensions) and all five random seeds (22, 32, 42, 52, 62). All values are produced by the unified closed-loop re-evaluation protocol described in the main text: every checkpoint is reloaded and re-run on the identical environment code, scenario set, and validity gates. The violation rate is the hard-constraint violation rate (spacing lower bound, TTC, jerk); car-following upper-bound breaches are gated separately in the validity check. All metrics are conditional statistics over the scenarios that passed the validity check, except for the Valid Scenario Ratio itself. The deterministic controllers (ACC, MPC, MPC-L, LQR, IDM) contain no random initialization and are therefore evaluated once under the same protocol.

S6. PER-SEED RAW VALUES OF THE ABLATION CONFIGURATIONS

Table S7 lists the per-seed raw values underlying every mean $\pm$ std entry of the two ablation groups of the main-text ablation table (the predictive-information group and the adaptive-mechanism group). All configurations were retrained as independent five-seed runs and re-evaluated under the same unified closed-loop protocol as Table S6, so the violation-rate column is the hard-constraint violation rate. The complete-model rows are independent replicas of the main-table PA-CSAC training run, not copies of it.

S7. PER-SEED RAW VALUES OF THE REWARD-COEFFICIENT SENSITIVITY CONFIGURATIONS

Table S8 lists the per-seed raw values underlying every mean $\pm$ std entry of the reward-coefficient sensitivity table in the main text. All nine configurations were retrained as independent runs on the complete five-seed set (22, 32,

TABLE S5
FULL PREDICTION-ERROR INJECTION GRID (MEAN $\pm$ STD OVER FIVE SEEDS 22, 32, 42, 52, 62 UNDER THE UNIFIED CLOSED-LOOP PROTOCOL)

Error Type	Fuel (L/100km)	Gap RMSE (m)	Hard Viol. Rate	Valid Scen. Ratio
Baseline	6.43 $\pm$ 1.22	21.14 $\pm$ 3.22	0.025 $\pm$ 0.016	0.92 $\pm$ 0.12
Bias injection p05	6.57 $\pm$ 1.26	21.12 $\pm$ 3.02	0.027 $\pm$ 0.015	0.92 $\pm$ 0.12
Bias injection p10	6.52 $\pm$ 1.00	21.08 $\pm$ 3.88	0.028 $\pm$ 0.014	0.91 $\pm$ 0.15
Bias injection n05	6.53 $\pm$ 1.27	21.10 $\pm$ 3.35	0.028 $\pm$ 0.017	0.89 $\pm$ 0.17
Bias injection n10	6.47 $\pm$ 1.32	22.08 $\pm$ 3.79	0.028 $\pm$ 0.018	0.92 $\pm$ 0.12
Residual injection $\times 0.5$	6.40 $\pm$ 1.17	21.58 $\pm$ 3.59	0.026 $\pm$ 0.013	0.92 $\pm$ 0.12
Residual injection $\times 1.3$	6.40 $\pm$ 1.22	21.01 $\pm$ 3.28	0.025 $\pm$ 0.016	0.92 $\pm$ 0.12
Residual injection $\times 1.6$	6.43 $\pm$ 1.08	21.53 $\pm$ 3.65	0.025 $\pm$ 0.014	0.92 $\pm$ 0.12
Residual injection $\times 2.0$	6.44 $\pm$ 1.12	21.13 $\pm$ 3.34	0.027 $\pm$ 0.013	0.92 $\pm$ 0.12
Residual injection $\times 3.0$	6.57 $\pm$ 1.25	21.29 $\pm$ 3.70	0.027 $\pm$ 0.014	0.92 $\pm$ 0.12
Mean replaced by persistence	6.57 $\pm$ 1.14	21.81 $\pm$ 3.48	0.027 $\pm$ 0.014	0.91 $\pm$ 0.15
Uncertainty injection $\times 0.5$	6.27 $\pm$ 1.14	22.92 $\pm$ 5.60	0.024 $\pm$ 0.015	0.89 $\pm$ 0.17
Uncertainty injection $\times 1.5$	7.03 $\pm$ 1.48	21.50 $\pm$ 3.07	0.030 $\pm$ 0.013	0.89 $\pm$ 0.17
Uncertainty injection $\times 2.0$	6.89 $\pm$ 1.16	20.14 $\pm$ 3.35	0.031 $\pm$ 0.017	0.92 $\pm$ 0.12
Combined injection mild	6.76 $\pm$ 1.23	21.21 $\pm$ 3.08	0.029 $\pm$ 0.016	0.92 $\pm$ 0.12
Combined injection standard	6.85 $\pm$ 1.18	21.04 $\pm$ 3.00	0.029 $\pm$ 0.015	0.93 $\pm$ 0.10
Combined injection severe	7.62 $\pm$ 1.75	20.36 $\pm$ 1.97	0.037 $\pm$ 0.021	0.88 $\pm$ 0.20

Note: p05/p10 indicate positive velocity biases of +0.5/+1.0 m/s, and n05/n10 indicate negative velocity biases of $-0.5/-1.0$ m/s; $\times 0.5-\times 3.0$ denote scaling factors for residual or uncertainty scales; the persistence row sets the predicted mean to the last observed leading speed at every horizon (residual scale zero) while keeping the learned dispersion channel; the safety metric is the hard-constraint violation rate (spacing lower bound, TTC, jerk), identical to the main comparison table; the combined configurations superimpose all three perturbation types with magnitudes ($\times 1.3$, +0.3 m/s, $\times 1.3$) for mild, ($\times 1.5$, +0.5 m/s, $\times 1.6$) for standard, and ($\times 2.0$, +1.0 m/s, $\times 2.0$) for severe.

42, 52, 62) under the 12,000-step screening budget declared there and evaluated on the same 17 NGSIM scenarios; the violation-rate column is the combined soft-constraint rate, consistent with the main-text sensitivity table. The three default ($\times 1.0$) rows retrain an identical configuration on the identical seed set as independent runs; their residual differences quantify training-run-to-training-run variability under the screening budget.

S8. ALL-SCENARIO UNCONDITIONAL STATISTICS

Table S9 reports the unconditional aggregation of the unified closed-loop re-evaluation: all 17 test scenarios enter every method’s mean with identical weight, and episodes failing the validity gate are retained with their partial-episode statistics rather than excluded. This aggregation removes the coverage asymmetry of the conditional statistics in the main comparison table, where each method is averaged over its own valid subset; the two aggregations are computed from one identical rollout set (the per-seed, per-scenario files of the re-evaluation pipeline), so their differences reflect the validity gate alone. Invalid episodes are identified by the same dual-channel gate as everywhere else (hard-constraint violation rate and upper-bound spacing-breach rate, each at most 0.20, plus gap RMSE at most 50 m and no collisions); their fuel and violation entries here are the statistics accumulated up to the point of evaluation and are therefore comparable across methods only in this flagged, unconditional sense.

Reading the two aggregations jointly preserves the ordering conclusions of the main text on a matched basis: PA-CSAC remains the only learning-based method that combines a fuel level at or below the strongest traditional controller with near-complete coverage, and its all-scenario hard-violation rate (0.023 ± 0.014) stays below those of DDPG, PPO, and PPO-Lagrangian; the still lower rates of SMORL (0.002), SAC (0.003), and TD3 and HRL (0.005) accompany the loose-following regime. The unconstrained off-policy baselines and the two published eco-driving strategies retain their characteristically low unconditional fuel under the loose-following regime documented in the main text, and the PPO pair remains above the ACC fuel level in this aggregation as well.

TABLE S6

PER-SEED RAW VALUES OF THE MAIN COMPARISON (LEARNING-BASED METHODS: FIVE RANDOM SEEDS; DETERMINISTIC BASELINES: SINGLE EVALUATION). THE PPO-LAG ROWS OF SEEDS 52 AND 62 COINCIDE AT THE DISPLAYED PRECISION; ACROSS THE SIX PERTURBED ACCOUNTING CONFIGURATIONS OF THE ROBUSTNESS REPLAY, THE TWO CHECKPOINTS' AGGREGATES COINCIDE OR DIFFER AT THE 10^{-4} – 10^{-3} LEVEL IN A CONFIGURATION-DEPENDENT PATTERN (E.G., CONDITIONAL GAP RMSE 16.9565 M VERSUS 16.9560 M), IDENTIFYING TWO NUMERICALLY DISTINCT POLICIES CONVERGED TO THE SAME OPERATING POINT.

Method	Seed	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Violation Rate	Valid Scen. Ratio
ACC	–	13.02	16.20	1.12	0.071	1.00
MPC	–	16.19	19.35	1.30	0.105	1.00
MPC-L	–	15.62	18.71	1.31	0.114	1.00
LQR	–	14.89	20.03	1.40	0.087	1.00
IDM	–	10.49	14.68	0.85	0.042	1.00
DDPG	22	4.72	32.52	0.19	0.003	0.59
	32	4.56	33.07	0.21	0.003	0.59
	42	4.77	34.65	0.20	0.003	0.59
	52	4.65	33.02	0.20	0.004	0.65
	62	16.18	14.57	1.27	0.149	0.53
TD3	22	4.99	34.16	0.21	0.007	0.71
	32	4.61	32.53	0.21	0.006	0.59
	42	4.48	34.25	0.22	0.005	0.65
	52	4.66	32.58	0.24	0.003	0.59
	62	4.67	36.58	0.19	0.003	0.59
SAC	22	4.96	35.25	0.19	0.002	0.53
	32	5.01	34.29	0.19	0.000	0.59
	42	5.24	36.66	0.22	0.003	0.53
	52	4.39	35.33	0.22	0.001	0.59
	62	4.66	34.41	0.16	0.001	0.65
PPO	22	16.27	15.86	1.27	0.160	0.71
	32	15.94	16.95	1.27	0.156	0.71
	42	14.22	17.91	1.10	0.138	0.76
	52	14.20	17.50	1.02	0.126	0.76
	62	15.19	16.95	1.20	0.151	0.53
PPO-Lag	22	14.90	16.57	1.21	0.156	0.82
	32	15.79	16.33	1.25	0.164	0.76
	42	32.65	45.34	0.39	0.000	0.06
	52	15.79	16.96	1.25	0.162	0.82
	62	15.79	16.96	1.25	0.162	0.82
SMORL	22	4.88	39.49	0.16	0.000	0.47
	32	3.74	35.29	0.18	0.000	0.47
	42	5.17	38.54	0.16	0.000	0.41
	52	5.99	37.16	0.17	0.000	0.35
	62	4.90	40.72	0.16	0.000	0.47
HRL	22	4.75	33.27	0.20	0.003	0.59
	32	4.43	34.77	0.19	0.004	0.65
	42	4.54	32.97	0.21	0.004	0.65
	52	4.51	32.55	0.21	0.005	0.65
	62	4.58	32.85	0.21	0.005	0.65
PA-CSAC	22	8.07	16.51	0.55	0.042	1.00
	32	6.38	19.89	0.42	0.026	0.94
	42	7.17	21.82	0.57	0.039	0.71
	52	5.20	22.25	0.40	0.008	1.00
	62	5.35	25.22	0.39	0.011	0.94

TABLE S7
PER-SEED RAW VALUES OF THE ABLATION CONFIGURATIONS (FIVE RANDOM SEEDS 22, 32, 42, 52, 62, UNIFIED CLOSED-LOOP PROTOCOL)

Configuration	Seed	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Violation Rate	Valid Scen. Ratio
Full model (prob. dist. + uncert.)	22	4.92	10.61	0.37	0.007	0.94
	32	7.01	15.17	0.49	0.036	1.00
	42	5.41	25.39	0.42	0.010	0.59
	52	5.89	20.63	0.48	0.018	1.00
	62	5.52	20.10	0.42	0.010	0.88
Mean prediction only	22	6.69	13.50	0.47	0.039	1.00
	32	4.76	19.28	0.32	0.004	0.88
	42	5.01	30.44	0.30	0.008	0.65
	52	7.95	19.00	0.62	0.030	1.00
	62	4.25	21.60	0.19	0.002	1.00
No future leading-vehicle info	22	8.28	14.18	0.62	0.063	1.00
	32	7.56	12.49	0.57	0.043	0.82
	42	7.09	21.93	0.65	0.042	0.65
	52	4.78	21.79	0.26	0.006	1.00
	62	4.97	21.48	0.36	0.018	1.00
Complete model (adaptive family)	22	6.57	14.59	0.48	0.038	1.00
	32	5.70	13.21	0.39	0.019	0.94
	42	7.12	19.19	0.63	0.037	0.71
	52	5.46	20.51	0.35	0.012	0.94
	62	4.40	24.67	0.20	0.005	1.00
Remove adaptive weights	22	6.22	16.27	0.35	0.018	1.00
	32	5.39	13.15	0.38	0.021	0.94
	42	5.89	26.63	0.44	0.018	0.71
	52	6.70	23.66	0.48	0.012	0.88
	62	5.85	25.68	0.31	0.016	1.00
Remove dynamic safe distance	22	6.68	16.60	0.46	0.039	1.00
	32	5.57	13.10	0.37	0.016	0.94
	42	8.42	22.30	0.74	0.056	0.76
	52	6.68	22.39	0.46	0.011	1.00
	62	4.39	31.12	0.19	0.002	1.00
Remove both	22	7.26	14.20	0.56	0.045	1.00
	32	6.23	15.74	0.43	0.027	0.88
	42	5.82	29.24	0.40	0.008	0.65
	52	4.22	24.29	0.23	0.007	1.00
	62	4.61	29.28	0.23	0.007	0.94

TABLE S8
PER-SEED RAW VALUES OF THE REWARD-COEFFICIENT SENSITIVITY CONFIGURATIONS (FIVE RANDOM SEEDS 22, 32, 42, 52, 62,
12,000-STEP SCREENING BUDGET)

Coefficient	Level	Seed	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Violation Rate	Valid Scen. Ratio
α_{far} (car-following)	$\times 0.5$	22	3.86	17.09	0.19	0.003	1.00
		32	5.14	16.18	0.40	0.014	0.94
		42	5.05	14.55	0.43	0.014	1.00
		52	5.75	14.90	0.50	0.018	1.00
		62	4.66	12.67	0.39	0.007	0.94
	$\times 1.0$	22	4.82	10.90	0.42	0.009	0.76
		32	5.36	14.40	0.45	0.017	0.82
		42	8.47	13.58	0.74	0.052	0.82
		52	4.79	14.55	0.35	0.007	0.94
		62	4.39	13.85	0.23	0.007	1.00
	$\times 1.5$	22	6.70	12.89	0.67	0.031	0.82
		32	4.44	12.58	0.44	0.004	0.82
		42	6.17	16.17	0.50	0.017	1.00
		52	5.45	14.12	0.47	0.017	1.00
		62	8.00	19.51	0.59	0.034	1.00
k_{uncert} (safety-uncertainty)	$\times 0.5$	22	4.16	14.20	0.37	0.006	1.00
		32	5.26	14.77	0.45	0.018	0.88
		42	7.34	14.85	0.63	0.015	0.88
		52	7.03	16.71	0.54	0.043	1.00
		62	4.62	13.55	0.35	0.006	0.88
	$\times 1.0$	22	5.78	15.16	0.48	0.024	1.00
		32	7.07	18.14	0.55	0.044	1.00
		42	4.88	16.67	0.39	0.008	1.00
		52	5.58	16.62	0.47	0.014	0.94
		62	4.15	15.50	0.23	0.005	1.00
	$\times 1.5$	22	5.46	12.53	0.51	0.017	0.94
		32	4.04	15.69	0.40	0.004	0.94
		42	9.87	15.26	0.86	0.063	0.88
		52	5.63	16.50	0.48	0.015	1.00
		62	7.77	15.18	0.60	0.036	1.00
Shield cost ($k_{\text{push}}^c, k_{\text{pull}}^c$)	$\times 0.5$	22	6.33	14.89	0.51	0.024	1.00
		32	7.57	14.40	0.61	0.032	0.94
		42	4.83	13.27	0.44	0.005	0.88
		52	7.83	14.76	0.60	0.049	1.00
		62	5.76	14.24	0.42	0.013	1.00
	$\times 1.0$	22	5.08	12.81	0.44	0.012	0.94
		32	6.51	14.49	0.61	0.041	0.82
		42	6.57	16.62	0.68	0.045	1.00
		52	11.39	18.50	0.84	0.086	0.94
		62	6.58	14.92	0.47	0.029	1.00
	$\times 1.5$	22	5.86	14.03	0.51	0.013	0.88
		32	5.07	14.21	0.45	0.011	0.88
		42	8.71	16.33	0.72	0.059	0.94
		52	8.23	16.14	0.62	0.035	1.00
		62	4.97	14.54	0.33	0.012	1.00

TABLE S9

ALL-SCENARIO UNCONDITIONAL STATISTICS OVER THE COMPLETE 17-EPIISODE TEST SET (LEARNING-BASED METHODS: MEAN $\pm$ STD OVER FIVE RANDOM SEEDS 22, 32, 42, 52, 62; DETERMINISTIC BASELINES: SINGLE EVALUATION; EVERY EPISODE OF EVERY METHOD ENTERS ITS MEAN WITH IDENTICAL WEIGHT, INVALID EPISODES INCLUDED)

Method	Fuel (L/100km)	Hard-Constr. Violation Rate
ACC	13.02	0.071
MPC	16.19	0.105
MPC-L	15.62	0.114
LQR	14.89	0.087
IDM	10.49	0.042
DDPG	6.20 ± 3.83	0.030 ± 0.056
TD3	4.45 ± 0.04	0.005 ± 0.001
SAC	4.59 ± 0.10	0.003 ± 0.001
PPO	15.32 ± 0.81	0.174 ± 0.018
PPO-Lagrangian	13.70 ± 4.64	0.140 ± 0.077
SMORL	4.26 ± 0.29	0.002 ± 0.000
HRL	4.45 ± 0.02	0.005 ± 0.001
PA-CSAC	6.19 ± 1.18	0.023 ± 0.014

S9. SPLIT-CONFORMAL RECALIBRATION OF THE DISPERSION CHANNEL

Table S10 documents the split-conformal recalibration analysis of the deployed predictor’s dispersion channel on the vehicle-level splits of the control-side dataset, at the three horizons the closed loop actually consumes ($t+1$, $t+3$, $t+5$). Conformity scores $s_i = |y_i - \mu_i|/\hat{\sigma}_i$ are computed on the validation split (held out from predictor training), and the finite-sample 95% quantile $\hat{q} = \lceil (n+1)(1 - \alpha) \rceil$ -th order statistic with $\alpha = 0.05$ defines the recalibrated half-width $\hat{q}\hat{\sigma}$; the deployed intervals use $1.96\hat{\sigma}$. The recalibrated quantiles stay within $\pm 9\%$ of the deployed level at every horizon, and applying them on the test split leaves PICP at 0.990–0.997 with MPIW shifts of -3.4% to $+8.6\%$, so the over-coverage reported in the main text is not removable by a marginal rescaling of the dispersion level.

TABLE S10

SPLIT-CONFORMAL RECALIBRATION OF THE DEPLOYED DISPERSION CHANNEL (VALIDATION SPLIT: CALIBRATION; TEST SPLIT: EVALUATION; CONFORMITY SCORE $|y - \mu|/\hat{\sigma}$; TARGET COVERAGE 95%)

Horizon	n_{calib}	n_{test}	PICP _{raw}	MPIW _{raw} (m/s)	$\hat{q}$	PICP _{cal}	MPIW _{cal} (m/s)	MPIW change
$t+1$	1104	1173	0.991	5.155	1.893	0.990	4.980	-3.4%
$t+3$	1072	1139	0.996	4.890	2.064	0.997	5.149	$+5.3\%$
$t+5$	1040	1105	0.984	5.279	2.128	0.994	5.733	$+8.6\%$

S10. PER-SEED DUAL-VARIABLE TRACES OF PPO-LAGRANGIAN

Table S11 tabulates the dual-variable traces logged during PPO-Lagrangian training (one dual update per 2,048-step epoch; the 60,000-step budget admits 30 epoch records with indices 0–29, i.e., 29 dual updates after the initial epoch), providing the numeric basis of the convergence analysis in the main text. The multiplier λ is sampled at five-epoch intervals together with its final value; the constraint-return estimate $\hat{J}_c$ is summarized by its training maximum and its final value. Four seeds rise monotonically to final values of 0.90–1.27 without a settling epoch; seed 42 recedes over epochs 5–10 (0.356 to 0.270) while $\hat{J}_c$ turns negative and then resumes climbing to 0.93. The training maximum of $\hat{J}_c$ reaches 10.6–23.7, i.e., 35–79 times the threshold $d = 0.30$, and the final estimates remain 18–43 times the threshold on every seed, so the projected dual ascent is never arrested within the budget.

S11. PER-SEED RAW VALUES OF THE SHIELD-OFF REPLAY

Table S12 lists the per-seed raw values of the shield-off replay: the five PA-CSAC checkpoints are re-evaluated with the kinematic safety shield disabled while the actuator saturation and rate limiter are retained, under the identical closed-loop re-evaluation protocol of the main comparison. Disabling the shield removes the execution-time guarantee that all other results in this paper take for granted, so the validity columns of this table do not carry deployment significance; they quantify the share of closed-loop availability that the runtime layer, rather than the learned policy, provides. The

TABLE S11
DUAL-VARIABLE TRACES OF PPO-LAGRANGIAN OVER FIVE RANDOM SEEDS (THRESHOLD $d = 0.30$; λ SAMPLED AT THE LISTED EPOCHS AND AT THE FINAL EPOCH 29; $\hat{J}_c$ SUMMARIZED BY ITS TRAINING MAXIMUM AND FINAL VALUE)

Seed	$\lambda_{(0)}$	$\lambda_{(5)}$	$\lambda_{(10)}$	$\lambda_{(15)}$	$\lambda_{(20)}$	$\lambda_{(25)}$	$\lambda_{(29)}$	$\max \hat{J}_c$	$\max \hat{J}_c/d$	$\hat{J}_{c,(29)}$
22	0.000	0.139	0.249	0.403	0.536	0.729	0.900	10.56	35.2	9.56
32	0.000	0.192	0.367	0.566	0.723	0.921	1.079	12.51	41.7	7.45
42	0.000	0.356	0.270	0.503	0.708	0.799	0.928	23.67	78.9	5.35
52	0.000	0.225	0.480	0.690	0.895	1.105	1.273	13.13	43.8	5.89
62	0.003	0.229	0.473	0.601	0.812	0.979	1.162	17.40	58.0	12.93

five-seed mean VSR drops from 0.92 to 0.235, and one seed loses every test scenario, while the rate-limiter share of the remaining interventions rises to 0.31.

TABLE S12
SHIELD-OFF REPLAY OF PA-CSAC OVER FIVE RANDOM SEEDS (KINEMATIC SHIELD DISABLED; ACTUATOR SATURATION AND RATE LIMITER RETAINED; CONDITIONAL STATISTICS OVER EACH SEED'S VALID EPISODES; SEED 42 RETAINS NO VALID EPISODE)

Seed	Valid Episodes	VSR	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Hard Viol. Rate
22	3	0.176	7.44	19.81	0.246	0.029
32	6	0.353	6.17	22.04	0.268	0.026
42	0	0.000	—	—	—	—
52	7	0.412	4.55	23.65	0.180	0.049
62	4	0.235	5.57	26.62	0.123	0.000
Mean $\pm$ std	4.0	0.235 $\pm$ 0.161	5.93 $\pm$ 1.21	23.03 $\pm$ 2.86	0.204 $\pm$ 0.066	0.026 $\pm$ 0.020

S12. PER-SCENARIO DETAIL OF THE SEED-42 EXCLUSIONS

Table S13 lists the per-scenario statistics of the five test episodes excluded by the validity gate for seed 42 of PA-CSAC, the only seed whose Valid Scenario Ratio falls below 0.94 in the main comparison. Every exclusion is triggered by the upper-bound spacing-breach channel alone or by that channel combined with the gap-RMSE bound; no excluded episode records a collision, a TTC violation, or a jerk-constraint violation, and the hard-constraint violation rate of each excluded episode stays below 0.015. On the twelve episodes this seed passes, the conditional statistics (fuel 7.17 L/100km, gap RMSE 21.8 m, hard-violation rate 0.039) lie within the range of the other four seeds, so the exclusions isolate a following-compactness failure on a specific scenario subset rather than a global performance deficit of the seed.

TABLE S13
PER-SCENARIO STATISTICS OF THE FIVE VALIDITY-GATE EXCLUSIONS OF PA-CSAC SEED 42 (EXCLUSION CHANNELS: UPPER = UPPER-BOUND SPACING-BREACH RATE ABOVE 0.20; GAP = GAP RMSE ABOVE 50 M; NO EXCLUDED EPISODE RECORDS A COLLISION, TTC VIOLATION, OR JERK-CONSTRAINT VIOLATION)

Scenario	Upper-Breach Rate	Gap RMSE (m)	Hard Viol. Rate	Exclusion Channels
9	0.529	54.21	0.014	upper, gap
10	0.329	53.64	0.000	upper, gap
13	0.429	45.71	0.000	upper
14	0.571	63.48	0.014	upper, gap
15	0.500	46.95	0.000	upper

S13. PER-SEED RAW VALUES OF THE CONSTRAINT-HANDLING SCREEN

Table S14 lists the per-seed raw values underlying the constraint-handling×embedding screen of the main text. The five variants were independently retrained on the complete five-seed set (22, 32, 42, 52, 62) under the full training budget and re-evaluated through the unified per-scenario pipeline of the main comparison; the violation-rate column therefore reports the same hard-constraint definition as the main tables, and the Valid Scenario Ratio applies the dual-threshold validity gate. Every entry is read from the per-seed summary of the unified replay

(reeval_comp_ablation_summary.csv), which is generated from the stored per-scenario rollout files; the aggregate mean $\pm$ std values of the main-text table are computed from the five seed values shown here.

TABLE S14
PER-SEED RAW VALUES OF THE CONSTRAINT-HANDLING $\times$ EMBEDDING SCREEN (FIVE SEEDS 22–62; UNIFIED RE-EVALUATION PROTOCOL; VIOLATION RATE IS THE HARD-CONSTRAINT RATE; THE NO-EMBEDDING VARIANTS SPAN THE FIXED-PENALTY AND NO-CONSTRAINT CONDITIONS)

Variant	Seed	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Hard Viol.	VSR
Full (penalty, embed.)	22	6.20	21.42	0.38	0.020	1.00
	32	5.94	23.55	0.35	0.016	1.00
	42	7.14	18.17	0.61	0.031	1.00
	52	9.29	19.30	0.82	0.073	0.65
	62	4.54	23.46	0.21	0.009	1.00
Lagrangian (embed.)	22	6.20	21.43	0.38	0.020	1.00
	32	5.94	23.54	0.35	0.016	1.00
	42	5.38	28.02	0.37	0.017	0.53
	52	8.27	20.17	0.72	0.046	0.71
	62	4.66	27.42	0.23	0.007	1.00
No cost constraint (embed.)	22	5.85	22.15	0.36	0.012	1.00
	32	6.20	22.90	0.37	0.019	0.94
	42	7.41	21.39	0.62	0.037	0.71
	52	8.63	18.51	0.79	0.051	0.71
	62	4.27	33.27	0.23	0.004	0.65
Fixed penalty (no embed.)	22	5.46	26.27	0.30	0.013	0.94
	32	6.10	14.19	0.38	0.020	0.94
	42	5.96	17.85	0.43	0.022	1.00
	52	6.44	19.57	0.52	0.025	0.94
	62	6.93	23.41	0.57	0.030	0.71
No constraint, no embedding	22	4.98	28.56	0.26	0.010	0.94
	32	6.62	14.27	0.50	0.018	0.94
	42	5.12	35.71	0.22	0.002	0.53
	52	6.30	23.78	0.53	0.026	0.71
	62	5.63	27.15	0.41	0.019	0.71

S14. EFFICIENCY-CONSTANT ROBUSTNESS OF THE FUEL ACCOUNTING

Table S15 reports the fuel accounting robustness replay referenced in the main text. All ten methods of the base comparison grid (the MPC-L, SMORL, and HRL extensions excepted) are re-scored from their stored checkpoints under six perturbed settings that span every efficiency constant of the equivalent-consumption model—the motor and regeneration efficiencies ($0.92 \rightarrow 0.85/0.96$ and $0.70 \rightarrow 0.55/0.85$), the engine efficiency ($0.35 \rightarrow 0.28/0.42$), and the electricity-to-fuel equivalence factor ($2.8 \rightarrow 2.24/3.36$)—leaving every other parameter untouched. The motor and regeneration constants enter both the fuel accounting and the battery SOC state included in the observation vector, so those replays re-run the closed loop rather than re-scoring fixed trajectories; the induced policy drift is nevertheless negligible—the Valid Scenario Ratio of every method is identical across the seven columns, and the gap RMSE of every method moves by at most 0.05 m. The engine efficiency and the equivalence factor are pure accounting constants, so their replays leave the closed-loop trajectories unchanged and the fuel differences isolate the sensitivity of the reported levels to the accounting constants. The fuel ordering of these ten methods is identical in all seven columns; across the six perturbed settings the PA-CSAC margin against IDM spans 3.62–4.72 L/100km and the margin against ACC spans 5.51–8.21 L/100km.

S15. PER-SEED RAW VALUES OF THE DISPERSION-SOURCE SUBSTITUTION

Table S16 lists the per-seed raw values of the dispersion-source substitution referenced in the robustness analysis of the main text. The learned, step-varying standard-deviation channel of the deployed predictor is replaced, at every horizon, by a single constant—the mean of the per-horizon residual standard deviations measured on the validation split, 1.43 m/s under the $t+1/t+3/t+5$ deployment convention—while the predicted mean trajectory and every other

TABLE S15
FUEL CONSUMPTION UNDER PERTURBED EFFICIENCY ACCOUNTING (L/100KM; LEARNING-BASED METHODS: FIVE-SEED CONDITIONAL MEAN; TRADITIONAL CONTROLLERS: DETERMINISTIC; VSR IDENTICAL ACROSS ALL COLUMNS; $\eta_{\text{MOT}}/\eta_{\text{REGEN}}$ COLUMNS RE-RUN THE CLOSED LOOP, THE REMAINING FOUR ARE PURE RE-ACCOUNTING)

Method	Deployed	$\eta_{\text{mot}}/\eta_{\text{regen}}$		η_{eng}		k_{eq}	
		0.85/0.55	0.96/0.85	0.28	0.42	2.24	3.36
TD3	4.68	5.07	4.49	4.68	4.68	3.75	5.62
SAC	4.85	5.25	4.65	4.85	4.85	3.88	5.82
PA-CSAC	6.43	6.83	6.23	6.85	6.16	5.48	7.39
DDPG	6.98	7.40	6.76	7.43	6.67	5.95	8.01
IDM	10.49	11.00	10.23	11.57	9.77	9.25	11.73
ACC	13.02	13.43	12.82	15.06	11.66	12.05	14.00
LQR	14.89	15.35	14.66	17.24	13.33	13.79	15.99
PPO	15.16	15.67	14.91	17.44	13.64	13.95	16.37
MPC	16.19	16.67	15.95	18.81	14.44	15.05	17.33
PPO-Lagrangian	18.99	19.91	18.52	20.93	17.69	16.74	21.23
PA-CSAC margin vs. IDM	4.06	4.17	4.00	4.72	3.62	3.77	4.34
PA-CSAC margin vs. ACC	6.59	6.60	6.59	8.21	5.51	6.57	6.61

pipeline component are retained. The five PA-CSAC checkpoints are then re-evaluated under the identical closed-loop protocol; no retraining is involved. The per-seed values of the learned-dispersion panel reproduce the PA-CSAC rows of Table S6.

TABLE S16
DISPERSION-SOURCE SUBSTITUTION OVER FIVE RANDOM SEEDS (CONSTANT $\hat{\sigma} = 1.43$ M/S VERSUS THE LEARNED CHANNEL; CONDITIONAL STATISTICS OVER EACH SEED'S VALID EPISODES)

Dispersion	Seed	VSR	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Hard Viol. Rate
Constant 1.43 m/s	22	1.00	7.79	16.58	0.53	0.043
	32	0.94	5.98	19.63	0.40	0.026
	42	0.65	7.70	20.64	0.62	0.040
	52	1.00	5.15	22.02	0.40	0.008
	62	0.94	5.48	24.64	0.42	0.010
Learned $\hat{\sigma}$	22	1.00	8.07	16.51	0.55	0.042
	32	0.94	6.38	19.89	0.42	0.026
	42	0.71	7.17	21.82	0.57	0.039
	52	1.00	5.20	22.25	0.40	0.008
	62	0.94	5.35	25.22	0.39	0.011
Constant, mean $\pm$ std		0.91 $\pm$ 0.15	6.42 $\pm$ 1.24	20.70 $\pm$ 2.97	0.47 $\pm$ 0.10	0.025 $\pm$ 0.016
Learned, mean $\pm$ std		0.92 $\pm$ 0.12	6.43 $\pm$ 1.22	21.14 $\pm$ 3.22	0.47 $\pm$ 0.09	0.025 $\pm$ 0.016

S16. SHIELD-OFF REPLAY OF THE LEARNING BASELINES

Table S17 extends the shield-off replay of Table S12 to the five learning baselines, completing the attribution matrix referenced in the main text. Each method's five stored checkpoints are re-evaluated with the kinematic shield disabled while the actuator saturation and rate limiter are retained, under the identical closed-loop protocol and without any retraining; the shielded rows reproduce the main comparison. The off-policy baselines lose 67–86% of their scenario validity and their conditional fuel collapses to the 2.6–2.8 L/100km coasting regime, whereas the on-policy pair shifts by at most 0.05 in VSR and 1.1% in conditional fuel; PA-CSAC combines the largest availability loss with an unchanged conditional violation rate on its retained scenarios.

S17. CLOSED-FORM EXPRESSIONS OF THE AUXILIARY REWARD TERMS

Table S18 collects the closed-form expressions of the ten auxiliary reward terms of the main text, together with the near-boundary safety correction Δr_s^{nb} , grouped by objective category. Within each term, the output magnitude is bounded through tanh saturation or a fixed small amplitude, so that the auxiliary terms regularize boundary behaviors

TABLE S17

SHIELD-OFF REPLAY OF THE FIVE LEARNING BASELINES (FIVE-SEED MEAN $\pm$ STD; SHIELDED ROWS REPRODUCE THE MAIN COMPARISON; KINEMATIC SHIELD DISABLED, ACTUATOR SATURATION AND RATE LIMITER RETAINED; PA-CSAC SHIELD-OFF ROW FROM TABLE S12)

Method	Setting	VSR	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Hard Viol. Rate
DDPG	Shielded	0.59 ± 0.04	6.98 ± 5.14	29.57 ± 8.42	0.42 ± 0.48	0.032 ± 0.065
	Shield-off	0.08 ± 0.07	2.75 ± 0.00	32.44 ± 3.89	0.001 ± 0.001	0.000 ± 0.000
TD3	Shielded	0.62 ± 0.05	4.68 ± 0.19	34.02 ± 1.65	0.22 ± 0.02	0.005 ± 0.002
	Shield-off	0.19 ± 0.05	2.74 ± 0.03	38.00 ± 2.51	0.003 ± 0.001	0.009 ± 0.015
SAC	Shielded	0.58 ± 0.05	4.85 ± 0.33	35.19 ± 0.95	0.20 ± 0.03	0.001 ± 0.001
	Shield-off	0.19 ± 0.05	2.58 ± 0.21	30.96 ± 8.35	0.005 ± 0.004	0.000 ± 0.000
PPO	Shielded	0.69 ± 0.10	15.16 ± 0.95	17.03 ± 0.77	1.17 ± 0.11	0.146 ± 0.014
	Shield-off	0.68 ± 0.10	15.25 ± 0.94	16.82 ± 1.17	1.17 ± 0.11	0.150 ± 0.013
PPO-Lagrangian	Shielded	0.66 ± 0.34	18.99 ± 7.65	22.43 ± 12.81	1.07 ± 0.38	0.129 ± 0.072
	Shield-off	0.61 ± 0.31	19.19 ± 7.54	22.08 ± 13.02	1.04 ± 0.44	0.130 ± 0.073
PA-CSAC	Shielded	0.92 ± 0.12	6.43 ± 1.22	21.14 ± 3.22	0.47 ± 0.09	0.025 ± 0.016
	Shield-off	0.24 ± 0.16	5.93 ± 1.21	23.03 ± 2.86	0.20 ± 0.07	0.026 ± 0.020

without overriding the action ranking induced by the primary objectives. The intermediate quantities appearing in the expressions are defined in Section S2, and the classification and values of all coefficients follow Table S2.

TABLE S18

CLOSED-FORM EXPRESSIONS OF THE AUXILIARY REWARD TERMS AND THE NEAR-BOUNDARY SAFETY CORRECTION, GROUPED BY OBJECTIVE CATEGORY

Category	Term	Expression
Car-following quality	r_{gap}	$-0.27 \tanh(e_{\Delta}/4.5) - 0.006 e_{\Delta} - 2 \times 10^{-4} e_{\Delta}^2$
	r_{vm}	$-0.18 \tanh(v_e - v_l /3.2) \cdot (0.35 + 0.65 \kappa_{\text{catch}})$
	r_{stop}	$-0.40 \cdot \mathbb{I}[v_l > 2.0 \wedge v_e < 0.5]$
	r_{catch}	$0.35 \max(f_{\text{gap}}, f_{\text{rec}}) \cdot f_v$
	in-band bonus	$+0.18 (1 - \Delta /3.2) \text{ for } \Delta \leq 3.2 \text{ m, added to } r_f$
Control smoothness and actuator feasibility	r_{bb}	$-0.12 \max(-a, 0) \cdot f_{\text{gap}} \cdot \max(f_v, 0)$
	r_{app}	$-0.12 \max(a, 0) \tanh(d_{\text{close}}/2.5) - 0.08 \tanh(d_{\text{close}}/2.0) \tanh(v_{\text{close}}/0.8)$
	r_{comf}	$-0.20 \tanh(j /1.4) - 0.12 \tanh(a /2.2)$
	r_{brake}	$-0.06 \max(-a, 0)$
Safety preservation	r_{shield}	$-0.12 \tanh(\Delta a_{\text{push}}/0.8) - 0.07 \tanh(\Delta a_{\text{pull}}/0.8)$
	Δr_s^{nb}	$-0.35 \tanh(n/0.8) \tanh((v_e - v_l)/0.6)$, active if $\Delta s - d_{\text{safe}} < 2.0 \text{ m}$ and $v_e - v_l > 1.0 \text{ m/s}$, with $n = (2.0 - (\Delta s - d_{\text{safe}}))_+$
Energy-efficient operation	r_{SOC}	$-0.14 \tanh(\text{SOC} - \text{SOC}^* /0.018)$, $\text{SOC}^* = 0.60$

Note: a is the commanded acceleration, j the jerk, and Δa_{push} and Δa_{pull} the downward and upward action modifications imposed by the shield; the intermediate quantities (Δ , d_{up} , e_{Δ} , κ_{catch} , f_{gap} , f_{rec} , f_v , d_{close} , v_{close}) are defined in Section S2, together with the units of all quantities. Every reward term is dimensionless (per-step reward units); the classification and values of all coefficients follow Table S2.

S18. COMPUTATIONAL COST OF THE COMPARED METHODS

Table S19 reports the computational cost of each method consolidated from the main text: the number of learned parameters, the per-step inference latency, and the training wall-clock time. The parameter count is obtained directly from the saved deployment checkpoints. The inference latency is the per-decision-step time of the controller call measured during the closed-loop evaluation; for PA-CSAC, this covers the probabilistic feature embedding and the policy forward pass, with the Transformer predictor being evaluated offline and its distributional outputs streamed as state features; for SMORL, it covers the receding-horizon trajectory optimization together with the safe-set evaluation; for HRL, execution invokes the lower-layer network alone; and for the deterministic traditional controllers, the dispersion reflects run-to-run timing noise. For the training time, each model-free method is run over instrumented segments of 4,000 and 8,000 steps under identical hardware; the steady-state rate is obtained by two-point differencing, which cancels one-time initialization costs, and is linearly extrapolated to the 60,000-step budget. The SMORL and HRL training entries are measured full-budget wall-clock times averaged over the five seeds and therefore include the periodic holdout evaluations of checkpoint selection, so they are conservative relative to the extrapolated entries. Because the training-time entries subsume different warm-start structures (the shared actor warm start of the model-free family

and the IDM-episode warm start of SMORL), these entries are order-of-magnitude estimates, and the relative comparison is unaffected. The traditional controllers are analytic or search-based and involve no learned parameters, so only their inference latency applies; the MPC and MPC-L latency entries support order-of-magnitude comparison only, and a paired same-session benchmark of the two (19.80 versus 19.75 ms) confirms that substituting the learned mean preview carries no systematic latency penalty.

TABLE S19
COMPUTATIONAL COST OF THE COMPARED METHODS. PARAMETERS ARE COUNTED FROM DEPLOYED CHECKPOINTS; INFERENCE LATENCY IS THE PER-STEP MEAN $\pm$ STD OVER FIVE SEEDS; TRAINING TIME IS LINEARLY EXTRAPOLATED FROM STEADY-STATE THROUGHPUT ON IDENTICAL HARDWARE.

Method	Params (M)	Latency (ms/step)	Rate (s/1k)	Time (h, extrap.)
ACC	–	0.03 ± 0.03	–	–
IDM	–	0.06 ± 0.06	–	–
MPC	–	24.95 ± 24.43	–	–
MPC-L	–	16.08 ± 2.91	–	–
LQR	–	0.69 ± 0.68	–	–
DDPG	0.18	0.62 ± 0.53	8.62	0.14
TD3	0.21	0.56 ± 0.33	11.60	0.19
SAC	0.21	0.58 ± 0.33	16.29	0.27
PPO	0.14	1.18 ± 1.21	1.94	0.03
PPO-Lagrangian	0.21	0.66 ± 0.07	0.99	0.02
SMORL	0.41	93.56 ± 5.76	113.07	1.88
HRL	0.35	0.37 ± 0.11	11.18	0.19
PA-CSAC (ours)	0.38	1.12 ± 0.97	79.19	1.32

S19. COMPLETE CONFIGURATIONS OF THE COMPARED BASELINES

This section consolidates the complete parameter settings of the twelve baseline methods and the shared training configuration, complementing the summary descriptions of the main text.

S19.1 Traditional and Optimal Controllers

The ACC baseline tracks a constant time-gap reference ($d_0 = 5.0$ m, $T = 1.2$ s) with a proportional–derivative controller ($k_p = 0.28$, $k_d = 0.65$); its reference gap takes the maximum of the time-gap distance and the dispersion-scaled dynamic safe distance ($0.9 \hat{\sigma}$). The IDM is used in its classical form with $v_0 = 33.0$ m/s, $d_0 = 5.0$ m, $T = 1.2$ s, $a_{\max} = b = 2.0$ m/s², and $\delta = 4$. The MPC baseline performs a receding-horizon beam search over a nine-level discrete acceleration set (-3.0 to 2.0 m/s²) with an 8-step horizon and 14 candidate beams, replanning at every control step; its stage-cost weights are gap 1.0, relative velocity 0.35, acceleration 0.02, jerk 0.08, and safety margin 10.0, following the priority hierarchy of the PA-CSAC reward and fixed for all scenarios and seeds. The MPC-L variant retains the beam-search structure, horizon, candidate count, stage-cost weights, and dispersion-scaled gap law of MPC unchanged and replaces only the horizon reference for the leading-vehicle speed—held at its observed value in MPC—with the learned mean preview consumed by the learning-based controllers (the $t+1/t+3/t+5$ predictive means, linearly interpolated over the 8-step horizon and held at the $t+5$ value beyond it), so that the MPC–MPC-L pair differs in information set alone. The LQR baseline uses a quadratic cost on gap error, relative velocity, and acceleration. Both MPC and LQR are designed on the same three-state longitudinal model (gap error, relative velocity, ego acceleration) with a first-order actuator lag ($\tau_a = 0.5$ s).

S19.2 Learning-Based Methods

The off-policy methods (PA-CSAC, DDPG, TD3, and SAC) use policy networks with three 256-unit hidden layers and ReLU activations, and their critic networks use two hidden layers of 256 and 128 units; PPO and PPO-Lagrangian use policy and value networks with two 256-unit hidden layers and tanh activations. The Adam optimizer is uniformly employed, with learning rates $\eta_\pi = 3 \times 10^{-4}$, $\eta_Q = 3 \times 10^{-4}$, and $\eta_p = 1 \times 10^{-3}$ for the policy network, value network, and probabilistic feature embedding network, respectively; the discount factor is $\gamma = 0.99$ and the target soft update rate $\tau = 0.005$. All model-free methods use a replay buffer of 400,000 transitions and a batch size of 256; the replay configurations of SMORL and HRL follow their source protocols as described below.

For the PPO-Lagrangian dual update, one epoch denotes a 2,048-step on-policy rollout followed by the update sweeps, and the dual variable is updated by $\lambda \leftarrow \max(0, \lambda + \eta_\lambda(\hat{J}_c - d))$ with dual learning rate $\eta_\lambda = 5 \times 10^{-3}$.

For exploration, DDPG and TD3 inject zero-mean Gaussian noise with initial scales of 0.15 and 0.10, decaying linearly over training and floored at 20% of the initial scale; PA-CSAC applies the same decaying schedule with a smaller initial scale (0.012, floor 0.001); SAC relies on its entropy-regularized stochastic policy. The temperature α in both SAC and PA-CSAC starts from 0.2 and is tuned automatically from the entropy condition with target entropy -1 ; the SAC baseline lets α evolve freely, whereas PA-CSAC projects α into $[0.02, 0.05]$ after every update.

The SMORL baseline follows the reference configuration of its source protocol: receding-horizon trajectory optimization over the 8-step horizon on the shared physics-based vehicle and energy model, a batch-constrained off-policy critic with (200, 100, 50)-unit hidden layers, a conditional variational autoencoder action sampler with latent dimension 5 and two 300-unit hidden layers drawing $n = 10$ candidate actions per step, and an autoregressive LSTM(50) safe-set density over the discretized state dimensions. It uses $\gamma = 0.995$, the Adam optimizer with learning rate 10^{-4} , a replay buffer of 200,000 transitions, batch size 256, soft target updates at $\tau = 10^{-3}$, and a fixed exploration rate $\varepsilon = 0.2$ on the discrete control lattice, and warm-starts its replay from 30 successful IDM episodes under its safe-set construction.

The HRL baseline applies deterministic policy gradients in both layers with network sizes, optimizers, and target-update rules identical to the DDPG baseline, so that the only structural difference from DDPG is the hierarchy itself. The upper layer plans (time-headway, SOC) goal trajectories at a macro period of ten control steps within goal bounds of $\text{THW} \in [0.8, 2.2]$ s and $\text{SOC} \in [0.45, 0.75]$, with the upper-layer discount factor set to γ^{10} and the per-period reward accumulated from the shared environment reward; the lower layer is a goal-conditioned deterministic policy executed at every control step through within-period goal interpolation, with the goal-tracking reference following the constant-time-headway spacing law ($d_0 = 5.0$ m) of the ACC baseline. Exploration adds decaying Gaussian noise to both the lower-layer action (initial scale 0.1, floored at 20% of the initial scale) and the goal reference.

S20. VEHICLE AND ENERGY-MODEL PARAMETER CONFIGURATION

Table S20 consolidates the core parameter settings used for the energy consumption and safety calculations of the main text: the longitudinal vehicle dynamics constants of the traction-force model, the parallel HEV component efficiencies, the battery limits, and the equivalence factor of the rule-based energy management strategy. The strategy prioritizes pure electric drive when the total power demand (wheel demand plus the auxiliary load P_{aux}) is below the motor power limit $P_{\text{mot}}^{\text{max}}$ and the SOC stays above its lower limit SOC_{min} ; otherwise, the power is distributed between the engine and the motor at a ratio of 0.8 : 0.2; during braking, regenerative recovery charges the battery up to $P_{\text{mot}}^{\text{max}}$ at the efficiency η_{regen} . The per-step equivalent consumption is floored at zero, so steps with a net regenerative surplus do not subtract from the cumulative total. Every evaluated controller is scored by this identical configuration, so the differences in equivalent energy consumption reported in the main text reflect differences in velocity trajectories only.

TABLE S20
CORE PARAMETER CONFIGURATION OF HYBRID ELECTRIC VEHICLE

Parameter	Symbol	Value	Unit
Curb weight	m	1500	kg
Aerodynamic drag coefficient	C_d	0.29	–
Frontal area	A	2.2	m ²
Rolling resistance coefficient	f_r	0.01	–
Rated battery capacity	E_{bat}	1.8	kWh
SOC lower limit	SOC_{min}	0.35	–
SOC upper limit	SOC_{max}	0.85	–
SOC target value	SOC^*	0.60	–
Engine efficiency	η_{eng}	0.35	–
Motor efficiency	η_{mot}	0.92	–
Regenerative braking efficiency	η_{regen}	0.70	–
Auxiliary power	P_{aux}	1.0	kW
Electric drive power upper limit	$P_{\text{mot}}^{\text{max}}$	15	kW
Equivalent fuel factor	k_{eq}	2.8	–

S21. REWARD-COEFFICIENT SENSITIVITY SCREENING

Table S21 consolidates the one-factor-at-a-time sensitivity screening summarized in the main text: the car-following weight α_{far} , the safety–uncertainty coupling coefficient k_{uncert} , and the shield-mismatch cost scale ($k_{\text{push}}^c, k_{\text{pull}}^c$) are each perturbed to $0.5\times$ and $1.5\times$ their default values, and every perturbed configuration is retrained on the same five seeds and the same 17 NGSIM test scenarios under the shortened per-run budget of 12,000 steps that keeps the 45-run grid tractable. Per-seed raw values are given in Table S8.

TABLE S21
SENSITIVITY ANALYSIS OF KEY REWARD COEFFICIENTS (MEAN $\pm$ STD OVER FIVE SEEDS UNDER THE SCREENING BUDGET; THE VIOLATION COLUMN REPORTS THE COMBINED SOFT-CONSTRAINT RATE; PER-SEED VALUES IN TABLE S8).

Coefficient	Level	Fuel (L/100km)	Gap RMSE (m)	Jerk RMSE (m/s ³)	Soft-Constraint Viol. Rate	Valid Scen. Ratio
α_{far} (car-following)	$\times 0.5$ (0.075)	4.89 $\pm$ 0.70	15.08 $\pm$ 1.69	0.38 $\pm$ 0.11	0.011 $\pm$ 0.006	0.98 $\pm$ 0.03
	$\times 1.0$ (0.150)	5.57 $\pm$ 1.66	13.46 $\pm$ 1.48	0.44 $\pm$ 0.19	0.018 $\pm$ 0.019	0.87 $\pm$ 0.10
	$\times 1.5$ (0.225)	6.15 $\pm$ 1.34	15.06 $\pm$ 2.86	0.54 $\pm$ 0.10	0.020 $\pm$ 0.012	0.93 $\pm$ 0.10
k_{uncert} (safety–uncertainty)	$\times 0.5$ (0.45)	5.68 $\pm$ 1.43	14.82 $\pm$ 1.18	0.47 $\pm$ 0.12	0.018 $\pm$ 0.015	0.93 $\pm$ 0.06
	$\times 1.0$ (0.90)	5.49 $\pm$ 1.09	16.42 $\pm$ 1.17	0.42 $\pm$ 0.12	0.019 $\pm$ 0.016	0.99 $\pm$ 0.03
	$\times 1.5$ (1.35)	6.56 $\pm$ 2.28	15.03 $\pm$ 1.50	0.57 $\pm$ 0.18	0.027 $\pm$ 0.023	0.95 $\pm$ 0.05
Shield cost scale ($k_{\text{push}}^c, k_{\text{pull}}^c$)	$\times 0.5$ (0.09, 0.04)	6.46 $\pm$ 1.25	14.31 $\pm$ 0.64	0.51 $\pm$ 0.09	0.025 $\pm$ 0.017	0.96 $\pm$ 0.05
	$\times 1.0$ (0.18, 0.08)	7.23 $\pm$ 2.41	15.47 $\pm$ 2.17	0.61 $\pm$ 0.17	0.042 $\pm$ 0.028	0.94 $\pm$ 0.07
	$\times 1.5$ (0.27, 0.12)	6.57 $\pm$ 1.78	15.05 $\pm$ 1.10	0.53 $\pm$ 0.15	0.026 $\pm$ 0.021	0.94 $\pm$ 0.06

Three observations follow from Table S21. First, none of the nine perturbed configurations collapses: the soft-constraint violation rate never exceeds 0.042, far below the 0.20 validity budget; the Valid Scenario Ratio remains at or above 0.87, and fuel consumption stays within 4.89–7.23 L/100km, below the levels of IDM, ACC, LQR, MPC, and PPO. Constraint satisfaction is thus enforced architecturally—by the cost channel and the kinematic shield—rather than by delicate reward tuning, and the reward coefficients mainly shape the efficiency–comfort trade-off within the feasible envelope. Second, the default $k_{\text{uncert}} = 0.9$ achieves the lowest fuel consumption, the lowest jerk, and the highest Valid Scenario Ratio within its group, supporting the calibration procedure of the main text. For the shield cost scale, raising it from the halved level to the default increases mean fuel consumption by 12% and jerk by 18% (computed from the unrounded seed-mean values), which is physically consistent with a stronger mismatch penalty keeping the policy farther from the intervention boundary at the cost of efficiency; the response saturates beyond the default, with the $1.5\times$ level falling back to 6.57 ± 1.78 L/100km. The halved and default shield scales differ by 0.77 L/100km, within one seed-level standard deviation (± 2.41 L/100km). The perturbations of α_{far} mainly redistribute the trade-off between tracking precision and energy consumption, with all three levels remaining operationally valid. Third, within every coefficient group, the mean shifts induced by the $\pm 50\%$ perturbations are of the same order as one seed-level standard deviation. The three default ($\times 1.0$) entries—identical configurations retrained independently—span up to 1.74 L/100km, indicating that seed-level stochasticity accounts for most of the within-group variation; the bounded perturbation response rules out knife-edge dependence on exact coefficient values.

S22. PROBABILISTIC CALIBRATION QUALITY

Table S22 reports the probability quality metrics of the two component-level variants of the prediction head (with and without the gating block); the corresponding reliability diagram of the calibrated predictive distribution on the test split is reported in the main text.

TABLE S22
PROBABILISTIC FORECASTING QUALITY EVALUATION RESULTS

Method	NLL	CRPS	PICP@95	MPIW@95 (m/s)
Transformer + GMH	0.0635	0.1384	0.9958	5.454
Transformer + TFG + GMH	0.0234	0.1276	1.000	5.606

Note: NLL is Negative Log-Likelihood, CRPS is Continuous Ranked Probability Score, PICP is 95% Prediction Interval Coverage Probability, MPIW is Mean Prediction Interval Width.

The probability quality evaluation shows that the combination of the temporal gating block and GMH reduces

both NLL and CRPS relative to the plain Transformer baseline. The realized PICP of 1.000 for the nominal 95% interval, combined with an MPIW of 5.606 m/s, indicates over-coverage: the calibrated intervals contain essentially every observed future speed, so the nominal 5% miss rate is never exercised on this test set.

The degree of inflation can be read directly from the two reported quantities: the interval half-width of 2.803 m/s corresponds, under the Gaussian interval convention, to an effective dispersion of $\bar{\sigma} = 5.606 / (2 \times 1.96) \approx 1.43$ m/s, roughly 1.7 times the horizon-aggregated RMSE of the deployed predictor (0.857 m/s, Table S1), so the dispersion signal consumed by the downstream safety distance exceeds the point-error scale by about this factor; in physical terms, this inflation corresponds to an average safety-margin increment of $k_{\text{uncert}} \bar{\sigma} \approx 1.3$ m in the dynamic safety distance of the main text. Since the uncertainty-aware safety distance consumes the predicted dispersion directly, systematically inflated intervals translate into a more conservative safety shield and a portion of the energy overhead reported in the comparative experiments of the main text. A contributing factor is the control-period sampling grid: the 1-s resampling of the 10-Hz NGSIM records coarsens speed transitions, and step-wise residuals on the coarsened grid are heavier-tailed than the underlying continuous dynamics. The split-conformal recalibration quantified in Section S9 shows that the over-coverage is not removable by a marginal rescaling of the dispersion level, so narrowing the intervals toward nominal 95% coverage calls for conditional, per-regime calibration.

S23. CHECKPOINT-INDEPENDENCE AUDIT OF THE PPO-LAGRANGIAN SEEDS 52 AND 62

The two converged PPO-Lagrangian seeds 52 and 62 report matching closed-loop aggregates in Table S6, and this section documents the audit that excludes a checkpoint-duplication artifact and locates the agreement at the behaviour level. The audit operates at four levels, each computed directly from the released artefacts.

At the file level, the SHA-256 digests of the five saved `ppo_lagrangian.pt` checkpoints are pairwise distinct (first 16 hexadecimal characters: seed 22, `d107901fd8220ddd`; seed 32, `1f62c8b590f5b6e3`; seed 42, `9ce9e6d616bc257f`; seed 52, `a26993157ac40c63`; seed 62, `93f30c5d1e066f1e`), so the seeds 52 and 62 rows cannot originate from one file loaded twice.

At the training-trajectory level, the logged episode-reward histories are distinct objects with different content: seed 52 records 893 episodes with first and last episode returns of -48.89 and -19.56 , seed 62 records 893 episodes with returns of -30.92 and -19.12 , and the two sequences differ episode by episode throughout training while converging to a common level, as expected of two independent runs of the same algorithm.

At the parameter level, Table S23 reports the pairwise Euclidean distances between the five actor networks over all weights and biases. All ten pairwise distances lie in the narrow band 24.19–24.42 against parameter norms of 17.12–17.23, i.e., relative distances of 1.41–1.42; in particular, the distance between seeds 52 and 62 (24.30) is indistinguishable from the distance between any other pair, so the two parameterizations are as mutually distant as any two independently trained seeds of the same architecture, which rules out weight sharing as well as file duplication.

TABLE S23
PAIRWISE ACTOR-NETWORK PARAMETER DISTANCES OF THE FIVE PPO-LAGRANGIAN SEEDS (DIAGONAL: PARAMETER NORM $\|\theta_\pi\|$;
OFF-DIAGONAL: EUCLIDEAN DISTANCE $\|\theta_\pi^{(i)} - \theta_\pi^{(j)}\|$ OVER ALL ACTOR WEIGHTS AND BIASES; ALL TEN PAIRWISE DISTANCES LIE IN
24.19–24.42, SO NO TWO SEEDS SHARE A PARAMETERIZATION)

Seed	22	32	42	52	62
22	17.12				
32	24.33	17.22			
42	24.38	24.38	17.23		
52	24.19	24.28	24.42	17.16	
62	24.28	24.31	24.36	24.30	17.19

At the behaviour level, Table S24 compares the per-scenario re-evaluation statistics of the seed pair 52–62 against a reference pair of independently behaving seeds (22–32). The 52–62 rollouts agree to within 7.3×10^{-5} L/100km in fuel, 4.3×10^{-3} m in gap RMSE, and 1.3×10^{-5} m/s³ in jerk RMSE (maxima over the 17 scenarios), with violation rates, actuator rate-limiter activation rates, shield activation rates, episode lengths, and validity flags exactly identical in every scenario, while the reference pair differs at whole-unit scale on the same quantities. The mechanism of the agreement is visible in the traces: the actuator rate limiter is active on 57.6% of the steps of both policies, so the deterministic action maps of the two parameterizations agree to within actuator saturation over effectively the whole episode, and the residual parameter differences survive only as the 10^{-5} -scale metric deviations of Table S24. The

agreement is therefore a behavioural convergence of two distinct parameterizations onto the same saturated braking-heavy action mapping—consistent with the dual-contraction interpretation of the main text—rather than a duplicated evaluation. The audit is reproducible with the released script `scripts/diag_ppolag_seed_distance.py`, which recomputes the digests, the distance matrix, and the agreement statistics from the checkpoints and per-scenario files.

TABLE S24
PER-SCENARIO ROLLOUT AGREEMENT BETWEEN PPO-LAGRANGIAN SEED PAIRS OVER THE 17 TEST SCENARIOS (THE SEED PAIR 52–62 MATCHES TO WITHIN 10^{-5} – 10^{-3} ON THE CONTINUOUS METRICS AND EXACTLY ON ALL DISCRETE FLAGS; THE REFERENCE PAIR 22–32 OF INDEPENDENTLY BEHAVING SEEDS DIFFERS AT WHOLE-UNIT SCALE)

Per-Scenario Metric	Seeds 52 vs 62		Seeds 22 vs 32 (reference)	
	max $ \Delta $	mean $ \Delta $	max $ \Delta $	mean $ \Delta $
Fuel (L/100km)	7.3×10^{-5}	1.4×10^{-5}	5.86	1.75
Gap RMSE (m)	4.3×10^{-3}	5.3×10^{-4}	17.82	3.91
Jerk RMSE (m/s ³)	1.3×10^{-5}	2.0×10^{-6}	3.2×10^{-1}	1.0×10^{-1}
Hard violation rate	0	0	0.114	0.042
Travelled distance (km)	3.3×10^{-5}	3.5×10^{-6}	3.5×10^{-2}	1.1×10^{-2}
Rate-limiter activation rate	0	0	0.243	0.087
Shield activation rate	0	0	0.029	0.008
Episode length (steps)	0	0	0	0
Validity flags	identical (17/17)		differ	

S24. CONDITIONAL BASIS OF THE SPEED- AND TIME-NORMALISED COMPARISON

Table S25 reports the conditional counterpart of the speed- and time-normalised comparison of the main text: each learning-based method is averaged over its own valid scenarios, so the rows are directly comparable with the conditional statistics of Table S6 and the main comparison table, while the unconditional table of the main text removes the coverage asymmetry. The two aggregations preserve the same ordering: the mean ego speed of PA-CSAC remains within the range spanned by the traditional controllers once its seed 42 is read jointly with its conditional-subset structure (the healthy seeds span 29.6–30.8 km/h on their valid subsets, and the five-seed conditional mean of 29.08 km/h is pulled down by seed 42’s 12-scenario subset at 24.93 km/h), the time-normalised fuel rate of PA-CSAC (1.65 ± 0.50 L/h) remains 37–63% below every traditional controller, and the unconstrained off-policy family combines the lowest fuel rates with the slowest ego speeds (23.99–24.45 km/h) and positive speed deficits relative to the lead vehicle, i.e., the loose-following regime documented in the main text. The two published eco-driving strategies occupy the same regime on this basis, with SMORL the slowest method of the comparison (19.29 km/h, a deficit of +0.78 km/h relative to the lead vehicle) and HRL at 24.69 km/h, so their low conditional fuel rates co-occur with the loose-following speed profile. Travel time is 70.0 s for every method on this basis by construction, because episodes that terminate early through the follow-loss bound fail the validity gate and are excluded from the conditional mean.

TABLE S25

SPEED- AND TIME-NORMALISED COMPARISON ON THE CONDITIONAL BASIS (LEARNING-BASED METHODS: MEAN $\pm$ STD OVER FIVE RANDOM SEEDS 22, 32, 42, 52, 62, EACH SEED AVERAGED OVER ITS OWN VALID SCENARIOS AS IN TABLE S6; DETERMINISTIC BASELINES: SINGLE EVALUATION; DEFINITIONS AS IN THE MAIN-TEXT TABLE; PPO-LAGRANGIAN SEED 42 CONTRIBUTES ITS SINGLE SURVIVING EPISODE, WHICH INFLATES ITS DISPERSION)

Method	Mean Ego Speed (km/h)	Travel Time (s)	Speed Deficit vs Lead (km/h)	Fuel Rate (L/h)	Fuel (L/100km)
ACC	30.60	70.0	-1.15	3.81	13.02
MPC	30.62	70.0	-1.17	4.51	16.19
MPC-L	30.77	70.0	-1.32	4.48	15.62
LQR	30.35	70.0	-0.90	4.21	14.89
IDM	30.77	70.0	-1.33	2.67	10.49
DDPG	24.09 $\pm$ 0.38	70.0	0.10 $\pm$ 0.54	1.30 $\pm$ 1.18	6.98 $\pm$ 5.14
TD3	24.37 $\pm$ 0.72	70.0	0.40 $\pm$ 0.14	0.79 $\pm$ 0.01	4.68 $\pm$ 0.19
SAC	24.45 $\pm$ 1.72	70.0	0.67 $\pm$ 0.41	0.77 $\pm$ 0.04	4.85 $\pm$ 0.33
PPO	28.52 $\pm$ 1.00	70.0	-1.05 $\pm$ 0.21	4.03 $\pm$ 0.16	15.16 $\pm$ 0.95
PPO-Lagrangian	23.99 $\pm$ 12.85	70.0	-0.36 $\pm$ 1.89	3.43 $\pm$ 1.73	18.99 $\pm$ 7.65
SMORL	19.29 $\pm$ 1.49	70.0	0.78 $\pm$ 0.23	0.60 $\pm$ 0.04	4.93 $\pm$ 0.81
HRL	24.69 $\pm$ 0.64	70.0	0.35 $\pm$ 0.06	0.79 $\pm$ 0.02	4.56 $\pm$ 0.12
PA-CSAC	29.08 $\pm$ 2.37	70.0	-0.74 $\pm$ 0.39	1.65 $\pm$ 0.50	6.43 $\pm$ 1.22